

Protocol FACS panel

FACS buffer: PBS Lonza + 2mM EDTA(Merck CBI 108418); 0,5% BSA (Roche 10735086001); 0,01% NaN₃ (Merck CBI 822335)

PBS: PBS Lonza BE17-516Q/12

IC kit: Intracellular Fixation & permeabilization buffer kit 88-882400 Ebioscience (125 ml) containing **Fix buffer**(Ebioscience 00-8222-49); **Permeabilization buffer** (Ebioscience 00-8333-56, 10x concentrated) dilute in MilliQ water

1. PBMC isolation → see leucosep PBMCs isolation protocol
2. Count cells using Beckman Coulter Cell counter.
 - i. Add 10 ml filtered Isotone in Accuvette
 - ii. Add 3 drops of ZAP-oGlobulinII
 - iii. Add 20 ul of cell suspension
 - iv. Mix gently and avoid airbubbles and incubate for 2'
 - v. Calculate cell concentration and resuspend cells in 2 ml RPMI⁺ medium
3. Divide cells over the different assays, in the following priority (I-V)
 - I. 2×10^6 PBMCs/ well (NUNC plate) → TLR panel FACS → continue with step 4 → 9.**
 - II. A cell suspension of 4×10^6 PBMCs/ml is made of the PBMCs for action III and IV.**
 - III. $5 \times 2 \times 10^6$ PBMCs/well (12 wells) + 5 stimulations with BFA (medium, Pam3CysK, LPS, R848, CpG) → cytokine panel FACS (1ml in total/well; single wells) → step 5**
 - IV. $5 \times 1 \times 10^6$ PBMCs/well (24 wells plate) + 5 stimulations (medium, Pam3CysK, LPS, R848, CpG) → sups 24h cytokine (0.5 ml in total /well ; in duplo if possible) → step 6 → step 8 → step 9**
 - V. Cryopreserve rest of cells (5×10^6 / vial) → step 7 & protocol cryopreservation of PBMCs**

When not enough cells are presented the first assay to be adapted is:

1. IV → from duplo to single wells per stimulation
2. V → no cryopreservation
3. IV → no 24h sups at all.

4. Start with staining TLR panel (1 person) → add 2×10^6 PBMCs/ well in NUNC plate and spin 400g for 3' at 4°C (continue at **step 9**)
5. 1-2 Persons start by plating cells $5 \times 2 \times 10^6$ PBMCs/well (12 wells plate) (**for intracellular cytokine staining**) and adding the five different stimulations (RPMI, Pam, LPS, R848 and CpG) **with BFA** and incubate for 3 hour (0,5 ml cells + 0,5 ml stimulation = 1ml)
6. 1-2 Persons start by plating cells $5 \times 1 \times 10^6$ PBMCs/well (24 wells plate) in duplo for all stimuli (if possible; otherwise priority all stimulations single wells) and adding the right stimulations (RPMI, Pam, LPS, R848 and CpG) **without BFA** for 24 hours (**for supernatant**); **write down time! (0.25 ml cells + 0.25ml stimulation = 0,5 ml)**
7. Cryopreserve left over PBMCs (see protocol cryopreservation)
8. Cytokine panel: After **3 hour of stimulation** spin the plate for 200 g for 5' and remove 500 ul supernatant (waste). Collect cells from the well in a 96 wells NUNC PP 500 ul plate with a **1ml pipette** by pipetting up and down carefully across the well (put your pipette on 200µl to prevent foaming etc.) **TIP: To collect the cells from the well pipet twice in the form of an X, twice as an + and three times around the side of the well**). Do this with 2-3 wells and check under the microscope. Pipet the loosened cells into a 96 wells NUNC PP 500 ul plate.

FACS protocol

9. For both FACS panels: Wash 1x by adding 200 ul FACS buffer and spin down the plate with cells 400g for 3' at 4°C
10. Discard the left supernatant by once! throwing the plate empty above the sink
11. Put 96wells plate on ice.
12. Resuspend and incubate the cells with 100µl in total (make 110 ul FACS buffer with **extracellular** surface markers antibodies + Fc block directly) in 96 well plate for the surface marker staining for 30 minutes on ice in *aluminium foil*. Antibody mixes see NOBLE_FACSpanels_finaldilutions excelsheet → prepared by Marloes day before
13. Wash 2x with 200 ul PBS and spin down the plate with cells 400g for 3' at 4°C
14. Stain cells with 100 ul aqua live/death FVD520 in PBS (1ul Live/death in 1000ul PBS per 10×10^6 PBMCs (make fresh)) (and conjugated streptavidin when needed). Antibody mixes see NOBLE_FACSpanels_finaldilutions excelsheet → prepared by Marloes day before
15. Incubate for 20' in the fridge
16. Add 180 ul FACS buffer to every well

17. Spin down for 3' minutes at 4°C and 400 g
18. Start Intracellular protocol Ebioscience:
19. 100 ul IC fixation buffer to each well and resuspend
20. Incubate in dark at RT! 30'
21. Add 100 ul 1x Perm buffer (cold)
22. Spin down for 3' minutes at 4°C and 400 g
23. Wash 2x by adding 200 ul 1x Perm Buffer and spin down for 3' minutes at RT and 400 g
24. Add the intracellular AB mix in permbuffer 100 ul/test and incubate for 20' in the dark in the fridge. Antibody mixes see NOBLE_FACSpanels_finaldilutions excelsheet → prepared by Marloes day before
25. Wash 2x by adding 200 ul 1x Perm Buffer and spin down for 3' minutes at RT and 400 g
26. **TLR panel** resuspend cells in **300 ul FACS buffer** and measure for **240s** with FACS CANTO II at medium flow rate, threshold 45.000
27. **Cytokine panel**: resuspend in **250 ul FACS buffer** and measure for **200s** with FACS Canto II at medium flow rate, threshold 45.000